ĐÁP SỐ

CHƯƠNG I: NHỮNG KHÁI NIỆM CƠ BẢN VỀ MẠCH ĐIỆN

1.1 J_S= -1A
1.2 P_{2A}= 72W
1.3 E = 24V
1.4 $P_{tải}$ = $P_{nguồn}$ = 208W
1.5 $P_{tải}$ = $P_{nguồn}$ = 4,5W
1.6 U = -14 V
1.7 U_0= 39V
1.8 U_1= 36V
1.9 U_S=-48V
1.10 J_1= 6A
1.11 U_S= 18V
1.12 U_0= 24V
1.14 U=177V
1.15 R_{td}= 12,5Ω
1.19 R_{td}= 22,5Ω
1.21 R_{td}= 2,249Ω
1.23 u=-3V
1.24 I_2=2A
1.25 I=1A
1.26 I_1=5A, I_2=-3A
1.29 i_0=-8mA
1.32 I_0= 9.23A
1.34 I_1=1,5A, I_2=0,5A
1.36 I_0= 16/3 mA
1.38 U_1= 35V, U_2=8V
1.39 I=0,527A
1.41 I=12 A
1.42 U_0=4V
1.43 I_1=3A
1.46 I_3=-1A
1.48 u_0= 6V
1.52 u_0= -2,57V
1.53 u_1= 1,2v, i=3,3 A
1.55 i_0= -4mA

CHƯƠNG II : CÁC PHƯƠNG PHÁP PHÂN TÍCH MẠCH

2.1 I = 0,37A

2.2 I = 0.611A

2.3 I = 3.88A

2.4 U = 8V

2.5 U_1 = -10V

2.6 I = 4A; U_1 = 0V

2.8 I_1 = 4A

2.9 I_1 = 0,01A; U = 10V; P_{ng} = 10,1W

2.10 I_1 = 1A; P_{ng} = 200W

2.11 U = 5V

2.12 U = 15V

2.13 I = 1A; V = 6V

2.14 I = 3A

2.15 I = 4A

2.16 $P_{4\Omega}$ = 64W

2.17 U = 8V

2.18 I = 1A

2.21 I = 3A

2.23 I = 2A

2.24 U_1= 5V

2.27 i= 1,2mA; u_0= 15V

2.29 i_1= -4A; i_2= 8A

2.31 u_1= 8V

2.34 i_1= 5A

2.37 i_0=8A

2.39 $P_{tải}$= $P_{nguồn}$= 126W

2.46 i_0= 3A

2.51 i_1= 5A; P_{4A}= -40W

2.55 i_2= 0,15A; u_1= -30V

2.57 i_0= 2mA

2.61 U_S= -45V

2.64 $P_{tải}$= $P_{nguồn}$= 2430W

2.65 $P_{tải}$= $P_{nguồn}$= 840W

2.66 $P_{tải}$= $P_{nguồn}$= 1775W

2.85 U=10V

2.90 R = 6Ω; P_{Rmax} = 8W

2.91 R = 4KΩ; P = 9mW

2.92 I = 2A

2.94

2.95 R=2,4 Ω, P = 0,1042W

2.96 I_R= 7

2.97 I_R = 1A

2.98 I_R = 3A

2.100 U_{ab}= 184,8V

2.104 I_{ng}= 60A

2.106 U_{ab}= -300V; I_{ng}= -30A

2.111 U_{ab}= 120V; I_{ng}= 40A

2.122 R = 16Ω

2.125 U_{ab}= -50V; $R_{tđ}$= 16Ω

CHƯƠNG III : MẠCH XÁC LẬP ĐIỀU HÒA

3.1 $Z_{tđ} = 1\angle 100^0$ Ω , $\dot{I} = 100\angle 0^0$ A, $\dot{I}_1 = 50\sqrt{2}\angle 45^0$ A; $\dot{I}_2 = 50\sqrt{2}\angle -45^0$ A, $\dot{V}_{12}$ = j100(V)

3.2 $\dot{I} = 32,16\angle 7^0$ (A) , $Z_{td} = 6,2\angle -7^0$ Ω, P = 3,2KW

3.3 $Z_i = 8,71\angle 31,48^0$ Ω , $\dot{I} = 11,48\angle -31,48^0$(A), $\dot{I}_1 = 11,2\angle -74,56^0$(A),

$\dot{I}_2 = 8,3\angle -34,62^0$(A), $\dot{U}_{12} = 41,39\angle 24,83^0$(V), $\dot{U}_{23} = 64,86\angle -15,56^0$(V)

3.4 $\dot{I}_1 = 4,64\angle 120,1^0$(A), $\dot{I}_2 = 17,4\angle 30,10$(A) $\dot{U}_{AB} = 11,6\angle -59,90$(V)

3.5 $Z_{tđ} = 7\angle 61^0$ Ω, $\dot{I} = 14,3\angle -61^0$(A), P_{tm}= 347W

3.6 $\dot{I} = 11,82\angle -7,1^0$ A

3.7 $P_{nguồn}$ =354W, $P_{5\Omega}$= 8,92W, $P_{3\Omega}$= 76,2W

$P_{5\Omega}$= 256,8W, $P_{2\Omega}$= 11,14W

3.10 V=$10\sqrt{2}$ V; V_1=10V ; V_2=20V ; V_3=30V

3.11 $u_0(t) = 20\sqrt{2}\cos(1000t - 135^0)$ V

3.12 $\dot{U}_{ab} = 28,59\angle 183^0 68$V

3.13 A= 18A, U_{ab}=25,2V

3.14 $\dot{I} = 33\angle -13^0$ A; $Z_{tđ} = 4,55\angle 58^0\Omega$

3.16 E= 53,53V

3.17 $\dot{I} = 10$A

3.18 $P_{nguồn}$ = 198W; $P_{5\Omega}$ = 85 W; $P_{3\Omega}$ =113W

3.19 $i(t) = 2\cos(t - 53^0 13)$ A; $i_1(t) = 2,83\cos(t + 171^0 87)$ A; $i_2(t) = 4,47\cos(t - 26^0 57)$ A

$P_{nguồn}$ = 6 W; $Q_{nguồn}$ = 8 Var

3.20 $\dot{I}_2 = 14{,}14\angle -45^0$ A; $\dot{I}_2 = 14{,}14\angle 45^0$ A ; $\dot{E} = 200{,}25\angle 2^0 86$V

$P_{nguồn}$ = 2099W; $Q_{nguồn}$ = - 1899Var; $S_{nguồn}$ = 2831VA.

3.21 $\dot{I}_2 = 4{,}47\angle -63^0 43$ A; $\dot{I}_1 = 4{,}47\angle 79^0 70$ A; $\dot{I} = 2{,}83\angle 8^0 13$ A

$P_{nguồn}$ = 140 W; $P_{3\Omega}$ = 60W; $P_{10\Omega}$ = 80W

3.22 P = 18W; $u_C = 4\sin(2t - 90^0)$ (V)

3.23 $\dot{I}_1 = 4\angle 90^0$; $\dot{I}_2 = 9{,}2\angle$-78; P= 900W

3.24 i(t)=4 cos(10t-53^0) A

3.27 $\dot{I}_1 = 17{,}6\angle 13{,}1^0$ A

3.30 A_2= 0,707A

3.31 i= 7sin(4t+29^0) A; Q= -17,14 Var; C = 0,04 F

3.32 r=15,75Ω; x_C =19,68Ω và r = 0,254Ω; x_C = 0,032Ω

3.33 A_1 chỉ 10A; A_3 chỉ 10A. R = 5Ω; x_C = 5Ω, x_L = 10Ω.

3.34 V= 40,41 V, P=346,8W, Z_c= -3j Ω

3.35 $\dot{I} = 1{,}01\angle 72{,}15^0$ A

3.38 u(t) = $5\sqrt{2}$ cos (6t – $36^0$87)V

3.39 u(t) = 9,6cos (4t – $53^0$13)V

3.40 Pe_1 = 11W; Pe_2 = 9,33W

3.41 u_C(t) = 1,57cos (t +18^0)V ; P= 0,38+ 0,24 W

3.42 $\dot{I}_1 = 1{,}19\angle 2^0$ A

3.51 $\dot{I}_1 = 10\angle 180^0$A

3.54 $\dot{I} = 3{,}67\angle -4{,}204^0$ A

3.56 i_1(t) = 7,59cos (4t + 108^0)A

3.59 $\dot{I}_2 = 6{,}12\angle -35^0$ A

3.62 $\dot{I} = 5{,}08\angle 6^0$ A

3.63 u_0(t)= 3.835cos(4t – 35.02°)V

3.64 i_0 (t) = 35.78 sin(1000t – 116.6°)A

3.65 $\dot{U}_0 = 16.64\angle 56.31^0$ V

3.70 u(t)=1,55sin(1000t -93^0)V

3.72 u(t) = 0,268cos (3000t - 155^0)V ; i(t) = 0,544cos (3000t -55^0)A
3.75 u_0(t) = 6,154cos(1000t +70^0)V

3.76 $\dot{I}_0 = 0$ A
3.77 u_1(t) = 2,48sin (2000t +$60,26^0$)V
3.78 R= 565,7 Ω
3.79 R_t= 14,23 Ω; P_m=0,24 W

3.83 $\dot{U}_{ab} = 350\angle 0^0$ V ; Z_{td}= 100+100j Ω

3.86 $\dot{U}_{ab} = 48 - 24j$ V ; Z_{td}= 20+10j Ω

3.88 $\dot{U}_{ab} = 80 + 60j$V ; $\dot{I}_{ng} = 10 + 20j$ A

3.89 $\dot{U}_{ab} = 37,95\angle -140^0$ V $Z_{td} = 7\angle -22^0\,\Omega$

3.91 $\dot{U}_{ab} = 7,35\angle 73^0$ V $Z_{td} = 4,32\angle -70^0\,\Omega$

3.94 $\dot{I}_0 = 1,971\angle -2^0$ A

3.114 $U_0 = \left(\frac{R_4}{R_3+R_4}\frac{R_1+R_2}{R_1}\right)U_2 - \left(\frac{R_2}{R_1}\right)U_1$; $\frac{R_4}{R_3+R_4} = \frac{R_2}{R_1+R_2}$

3.115 $U_0 = -R_0\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)$

3.117 U= 4V
3.118 I= 6mA
3.119 I = 0,5 mA; U_2= -1,5 V
3.120 R= 6/5Ω
3.121 U_0 = -4V
3.122 I = 1,5A
3.123 I= -0,25A
3.124 u_1=8V
3.127 U_0= 10V
3.128 U_0= -14V
3.129 U_0= -4V
3.130 U_0= -12V
3.131 U_0= -4V
3.136 u(t) = $4\sqrt{2}$ cos (5t +135^0)V

3.138 $\dot{I}_0 = 0,762\angle 30,1^0$ mA;

3.140 $\dot{U}_0 = -29,2\angle -69,4^0$ V

3.142 $\dot{U}_0 = 2{,}64\angle -165^0$ V

3.145 u(t) = $2{,}5\sqrt{2}\cos(20t + 135^0)$V

3.149 $\dot{U}_{ab} = 141.81\angle 23{,}2^0$ V $\dot{I}_{ng} = 20\sqrt{2}\angle -45^0$ A

3.151 $\dot{U}_0 = 21\angle -180^0$ V